

Psychometric Platform — Implementation Blueprint & Project Plan

Complete, Implementation-Ready Architecture Across All 5 Execution Phases

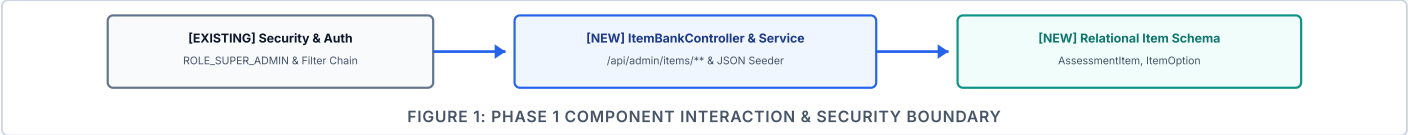
Status: Target Build Plan (Post-Auth Migration)

Spec Reference: PSYCHOMETRIC_EXAM_AND_REPORTING_SPECIFICATION_EN.pdf

Core Stack: Spring Boot 3.4+, Redis, MySQL/PostgreSQL

1 Phase 1: Item Bank & Data Model

1. **Overview:** Phase 1 establishes the psychometric question repository, trait mappings, and scoring benchmark keys for all 4 batteries (140 Personality PQ10, 16 SJT scenarios, 60 Derailers, 42 Cognitive items). It creates the authoritative question bank required by candidate testing and scoring engines.
2. **Component Architecture & Existing Module Connections:**



3. UML Class / Entity Architecture:

AssessmentItem (Entity: assessment_items) - id: Long «@Id, IDENTITY» - battery: AssessmentBattery «Enum: PQ10, SJT, DERAILERS, GCAT» - trait: CompetencyTrait «Enum: 10 PQ10, 3 Derailers, 3 GCAT» - itemCode: String «unique, nullable=false» - promptEn / promptAr: String «TEXT» - questionType: QuestionType «LIKERT_5, RANKING_4, MCQ» - idealTarget: Integer «1..5, null for SJT» - isReverseKeyed: boolean «default=false» - isQuickMode: boolean «default=false» - options: List<ItemOption> «@OneToMany, CASCADE=ALL»	ItemOption (Entity: item_options) - id: Long «@Id, IDENTITY» - item: AssessmentItem «@ManyToOne, FK: item_id» - optionKey: String «"A", "B", "C", "D"» - textEn / textAr: String «TEXT» - expertRank: Integer «1..4 (for SJT Kendall-tau)» - isCorrect: Boolean «true/false (for GCAT MCQs)» - displayOrder: int «1..4»
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4. Endpoint Design:

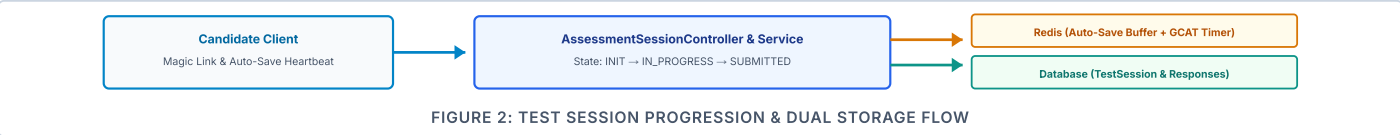
Method & Path	Purpose	Auth / Role	Request Body / Params	Response Shape	Validation & Business Rules
GET /api/admin/items	List items with filters	ROLE_ADMIN	Params: battery, trait, mode, page, size	200 OK Page<ItemResponse>	Includes ideal target keys for authorized admin review.
POST /api/admin/items	Create new assessment item	ROLE_SUPER_ADMIN	ItemCreateRequest (battery, trait, prompts, target, options)	201 CREATED ItemResponse	Validates Likert target between 1 and 5; validates SJT options have unique ranks 1..4.
POST /api/admin/items/seed	Seed 258 benchmark items	ROLE_SUPER_ADMIN	None (Reads embedded JSON seed dataset)	200 OK SeedSummaryResponse	Idempotent; populates 140 PQ10, 16 SJT, 60 Derailer, 42 GCAT questions.

5. **Step-by-Step Mechanism:** (1) Super Admin calls seed endpoint; (2) Service reads benchmark item JSON; (3) Validates 140 PQ10 items (10 traits x 14 items, balanced direct/reverse keys); (4) Validates 16 SJT scenarios with 4 options ranked 1–4; (5) Validates 42 GCAT items (14 Verbal, 14 Numerical, 14 Abstract); (6) Persists items in an atomic database transaction.
6. **Preconditions:** Security and database persistence layers active.
7. **Definition of Done:** All 258 benchmark items seeded in DB; automated tests verify exact target keys (T_i), reverse flags, and SJT rank sequences.

2 Phase 2: Test Session & Response Collection

1. **Overview:** Phase 2 implements test session state machine management, secure item delivery (stripping scoring keys from client responses), real-time background answer persistence to Redis, and authoritative server-side timing for the 20-minute GCAT constraint.

2. Component Architecture:



3. UML Class / Entity Architecture:

TestSession (Entity: test_sessions) - id: Long «@Id, IDENTITY» - sessionToken: String «unique, indexed» - candidate: User «@ManyToOne, FK: candidate_id» - examMode: ExamMode «FULL (258), QUICK (78)» - state: SessionState «INIT, IN_PROGRESS, SUBMITTED, SCORED» - startTime / submitTime: Instant - gcatStartTime: Instant «Server UTC start» - gcatTimeLimitSeconds: Integer «1200 (20 min)» - responses: List<CandidateResponse> «@OneToMany»	CandidateResponse (Entity: candidate_responses) - id: Long «@Id, IDENTITY» - session: TestSession «@ManyToOne, FK: session_id» - item: AssessmentItem «@ManyToOne, FK: item_id» - selectedLikert: Integer «1..5 (for PQ10 & Derailers)» - rankingOrder: List<String> «["B","A","D","C"] for SJT» - selectedOption: String «"C" for GCAT MCQs» - responseTimeMs: Long - submittedAt: Instant
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4. Endpoint Design:

Method & Path	Purpose	Auth / Role	Request Body / Params	Response Shape	Validation & Business Rules
POST /api/admin/sessions	Provision test for candidate	ROLE_ADMIN	SessionCreateRequest (candidateId, examMode)	201 CREATED SessionSummaryResponse	Creates session in INITIALIZED state; generates cryptographically secure session token.
GET /api/assessment/session/{token}/items	Deliver sanitized item bank	Session Token	Path: token	200 OK List<SanitizedItemResponse>	Strips idealTarget, expertRank, and isCorrect from client payload.
POST /api/assessment/session/{token}/auto-save	Persist in-flight answers	Session Token	AnswerBatchRequest (list of item answers)	200 OK { savedCount: int }	Writes immediately to Redis buffer; syncs to DB asynchronously.
POST /api/assessment/session/{token}/gcat/start	Start authoritative GCAT timer	Session Token	None	200 OK { timeLimitSeconds: 1200 }	Records server UTC timestamp; rejects restarts if already initiated.
POST /api/assessment/session/{token}/submit	Finalize assessment submission	Session Token	None	200 OK { status: "SUBMITTED" }	Locks session state to SUBMITTED; enforces GCAT time limit (20 min + 30s grace); triggers scoring.

5. **Step-by-Step Mechanism:** (1) Candidate opens assessment via secure magic link token; (2) Client fetches sanitized items; (3) As candidate progresses, selections auto-save to Redis; (4) Upon reaching GCAT, backend starts 20-min countdown; (5) Submission locks session to SUBMITTED state.

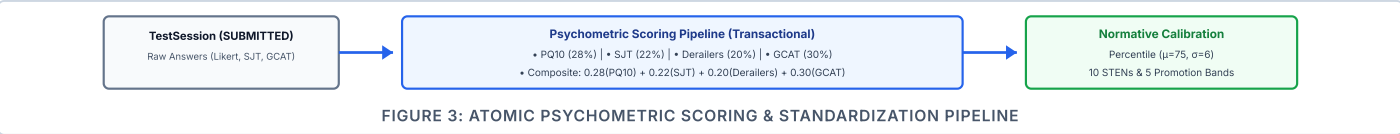
6. **Preconditions:** Phase 1 Item Bank active and populated.

7. **Definition of Done:** Zero scoring keys leak in API payloads; auto-save successfully recovers state on simulated browser crash; server enforces GCAT 20-minute cutoff.

3 Phase 3: Scoring Engine & Normative Calibration

1. **Overview:** The intelligence core of the platform. Implements the 4 exact psychometric grading algorithms (Likert absolute distance, Kendall-tau pairwise rank concordance, Derailer risk inversion, Cognitive accuracy), synthesizes the weighted composite score (28/22/20/30), and executes normative logistic percentile calibration and promotion readiness band mapping.

2. Component Architecture:



3. UML Class / Entity Architecture:

AssessmentScore (Entity: assessment_scores)

- id: Long «@Id, IDENTITY»
- session: TestSession «@OneToOne, FK: session_id»
- personalityScorePct: Double «Weight: 28%»
- sjtScorePct: Double «Weight: 22%»
- derailersEffectiveScorePct: Double «Weight: 20%»
- cognitiveScorePct: Double «Weight: 30%»
- compositeScorePct: Double «0.0 .. 100.0»
- percentileRank: Integer «1 .. 99»
- stanine: Integer «1 .. 9»
- promotionBand: ReadinessBand «EXCELLENT, STRONG, ...»
- traitScores: List<TraitScore> «@OneToMany»

TraitScore (Entity: trait_scores)

- id: Long «@Id, IDENTITY»
- assessmentScore: AssessmentScore «@ManyToOne»
- trait: CompetencyTrait «10 PQ10 Traits»
- rawScore: Double
- scorePct: Double
- stenScore: Integer «1 .. 10 (Mean 5.5, SD 2.0)»

4. Mathematical Scoring Formulas & Logic:

- PQ10 Distance:** $Distance_i = |A_i - T_i|$; $Points_i = 4 - Distance_i$; $PQ10\% = (Sum\ Points_i / (N \times 4)) \times 100$.
- SJT Concordance:** For all 6 pairs per scenario, award 1 if relative order matches expert consensus; $SJT\% = (Concordant\ Pairs / Total\ Pairs) \times 100$.
- Derailer Inversion:** $Risk\ Points_i = A_i - 1$; $Risk\% = (Sum\ Risk\ Points_i / (N \times 4)) \times 100$; $Effective\ Score\% = 100\% - Risk\%$.
- Composite Score:** $Composite = 0.28(PQ10\%) + 0.22(SJT\%) + 0.20(Derailers\%) + 0.30(GCAT\%)$.
- Percentile & Bands:** $Percentile = round[100 / (1 + exp(-(Score - 75.0) / 6.0))]$; $STEN = clamp(1, 10, round(5.5 + 2.0 \times (Trait\% - \mu) / \sigma))$. Promotion Readiness: Excellent (90–100%), Strong (80–89%), Acceptable (70–79%), Foundational Advanced (50–69%), Foundational (<50%).

5. Endpoint Design:

Method & Path	Purpose	Auth / Role	Request Body / Params	Response Shape	Validation & Business Rules
GET /api/admin/scores/{sessionId}	Fetch complete score dossier	ROLE_ADMIN	Path: sessionId	200 OK ScoreBreakdownResponse	Returns 4 battery percentages, composite %, percentile, STENs, and readiness band.
POST /api/admin/scores/{sessionId}/recalculate	Re-run scoring pipeline	ROLE_SUPER_ADMIN	Path: sessionId	200 OK ScoreBreakdownResponse	Re-grades raw answers against updated target keys/norms within an atomic transaction.

6. **Preconditions:** Phase 2 response storage functional.

7. **Definition of Done:** 100% test coverage over scoring algorithms matching hand-calculated spec benchmark test cases (e.g. Worked Example: 85.0% PQ10, 83.33% SJT, 91.67% Derailers, 88.10% GCAT yields 86.9% Composite → 88th Percentile → "Strong" Band).

4 Phase 4: PDF Report Generation Engine

1. **Overview:** Constructs the formal 5-Page Executive Leadership Dossier asynchronously upon test completion. Generates vector SVG radar charts, risk heatmaps, computes an immutable SHA-256 session signature, and stores the PDF in encrypted cloud storage with pre-signed retrieval.

2. 5-Page Executive Dossier Structure & Endpoints:

Dossier Page Architecture

- Page 1:** Executive Brief, Hero Score (0–100%), Percentile & Readiness Badges, Validity Index.
- Page 2:** PQ10 10-Axis Vector Radar Chart, STEN Scores (1–10), Top 3 Strengths & Development Needs.
- Page 3:** SJT Decision Style, Crisis & Conflict Efficacy, Scenario Consistency Metric.
- Page 4:** 3-Tier Derailer Heatmap (Green/Yellow/Red) & Cognitive Speed vs. Accuracy Diagnostic.
- Page 5:** Individual Development Plan (Start/Stop/Continue Matrix), Governance Sign-Off.

Endpoints (Report Module)

GET /api/admin/reports/{id}/download	ROLE_ADMIN	Returns 15-min pre-signed download URL.
GET /api/admin/reports/{id}/preview	ROLE_ADMIN	Returns HTML/SVG web preview of dossier.
GET /api/reports/verify/{hash}	Public	Validates SHA-256 cryptographic audit signature.

3. **Definition of Done:** Vector PDF compiles in <3.5s; 300+ DPI crisp vector charts; SHA-256 hash mathematically verifiable against candidate ID and score record.

5 Phase 5: Frontend, Redis Optimization & Scale

1. **Overview:** Completes candidate test-taking SPA (Likert matrices, drag-and-drop SJT ranker, Dual-Clock GCAT runner), admin dossier dashboard, Redis write-behind caching, rate limiting, and CDN static asset caching.

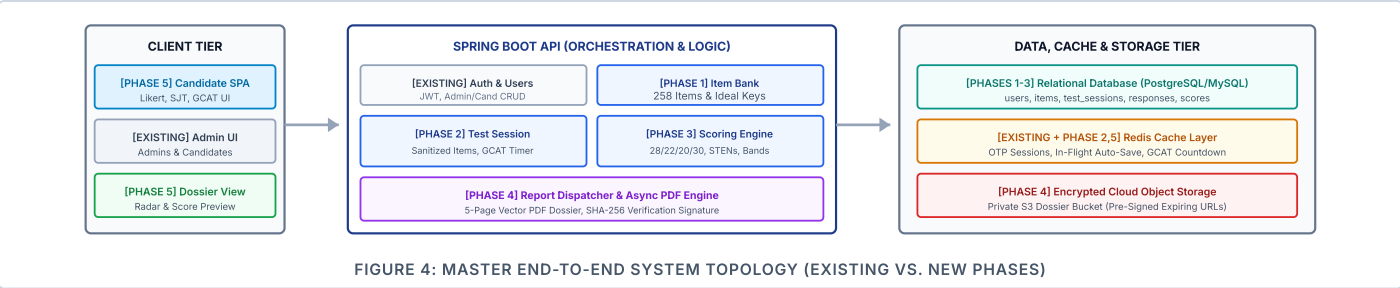
2. Frontend & Scale Architecture:

Candidate SPA Features 5-point interactive Likert matrix with smooth keyboard navigation. - Drag-and-drop 4-option SJT ranking with instant conflict resolution. - Dual-Clock GCAT runner with visual 5-min/1-min warning overlays. - Sub-second auto-save heartbeat with offline sync recovery.	Scale & Security Hardening - Token-bucket rate limiter on submission endpoints (max 10 req/min). - Static assets & report branding served via global Edge CDN. - Redis write-behind buffer protecting DB under 1000+ concurrent sessions. - Strict CORS and Content Security Policy (CSP) enforcement.
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3. **Definition of Done:** End-to-end candidate test flow functions seamlessly across mobile and desktop; zero progress loss during network disruption; load tested to 1,000 concurrent submissions.

6 Master System Architecture Diagram

Consolidated top-to-bottom architecture showing the entire platform once all 5 phases are completed:



7 Sequencing Rules & Project Risk Notes

Strict Implementation Sequencing Rules

- Rule 1:** Phase 2 (Session Engine) must NOT start until Phase 1 Item Bank seed tests pass 100%.
- Rule 2:** Phase 3 (Scoring Engine) unit tests must be validated against hand-calculated values before Phase 4 begins. Never build PDF templates against mock scores.
- Rule 3:** Phase 4 PDF Engine must run asynchronously to prevent blocking candidate submit responses.

Confirmed Normative Assumptions

- Percentile Distribution:** Logistic curve parameters verified at $\mu = 75.0$, $\sigma = 6.0$.
- Section Weighting:** Exact weights verified at PQ10 (28%), SJT (22%), Derailers (20%), GCAT (30%).
- Readiness Bands:** Excellent ($\geq 90\%$), Strong ($\geq 80\%$), Acceptable ($\geq 70\%$), Foundational Adv ($\geq 50\%$), Foundational ($< 50\%$).